Cpts 317. 9/28.

HW Prob 2.(2).

$$S \equiv aaaaa$$

---

all n s.t. $n \bmod 3 = 0$ or

$n \bmod 4 = 2$.

$$(aaa)^* + (aaaa)^* aa.$$

HW Prob 5.

Prove: $L^{-\Sigma'}$ is reg if L is reg.

Proof. 1.

2.

3.

I.H.

4.1

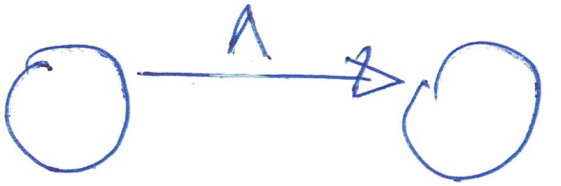
4.2

4.3.

d'b: if L_1 and L_2 are reg,
and $L_1^{-\Sigma'}$ & $L_2^{-\Sigma'}$ are reg, then,

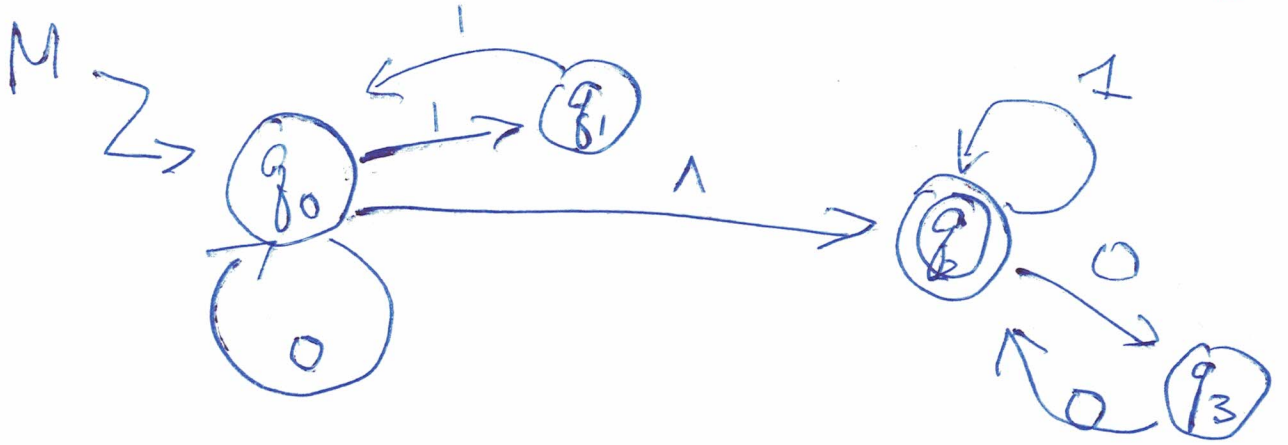
Λ-NFA (ε-NFA).

Λ-NFA = NFA + Λ-transitions



Example: Draw an Λ-NFA to accept

$$(11+0)^*(1+00)^*.$$



1100 is accepted by M since

$$\rightarrow q_0 \xrightarrow{1} q_1 \xrightarrow{1} q_0 \xrightarrow{1} q_2 \xrightarrow{0} q_3 \xrightarrow{0} q_2 \checkmark$$

$(11 \cdot 00 = 1100)$

L(M) is the set of all words accepted by M.

---

101 is not accepted by M.

110 ~~is not at~~ is accepted by M:

$$\rightarrow q_0 \xrightarrow{1} q_1 \xrightarrow{1} q_0 \xrightarrow{0} q_0 \xrightarrow{1} q_2 \checkmark$$

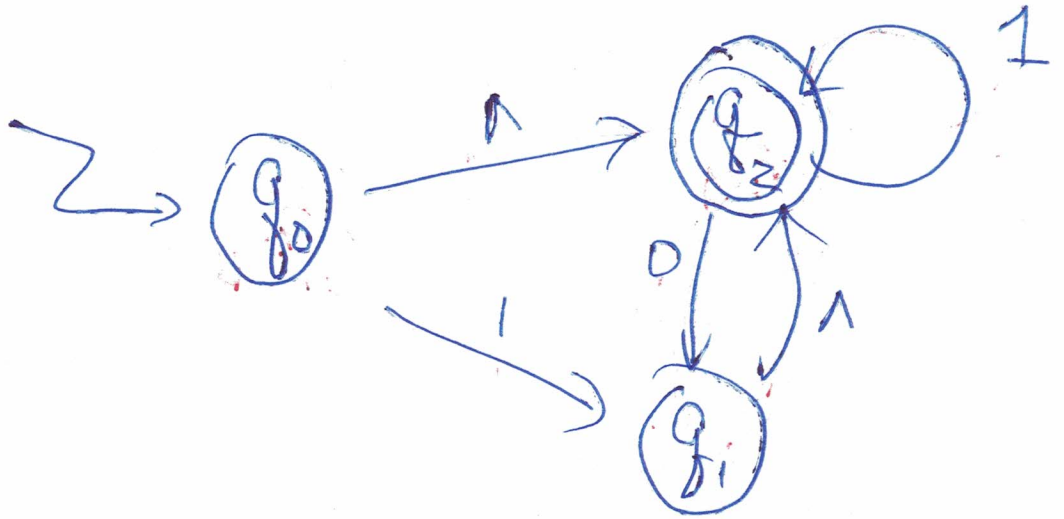
So far,

1. Cartesian Product Construction; (for "$\cap$")
2. Subset construction (NFA $\rightarrow$ DFA)
3. Complement (hw!)
4. Λ-elimination (today)

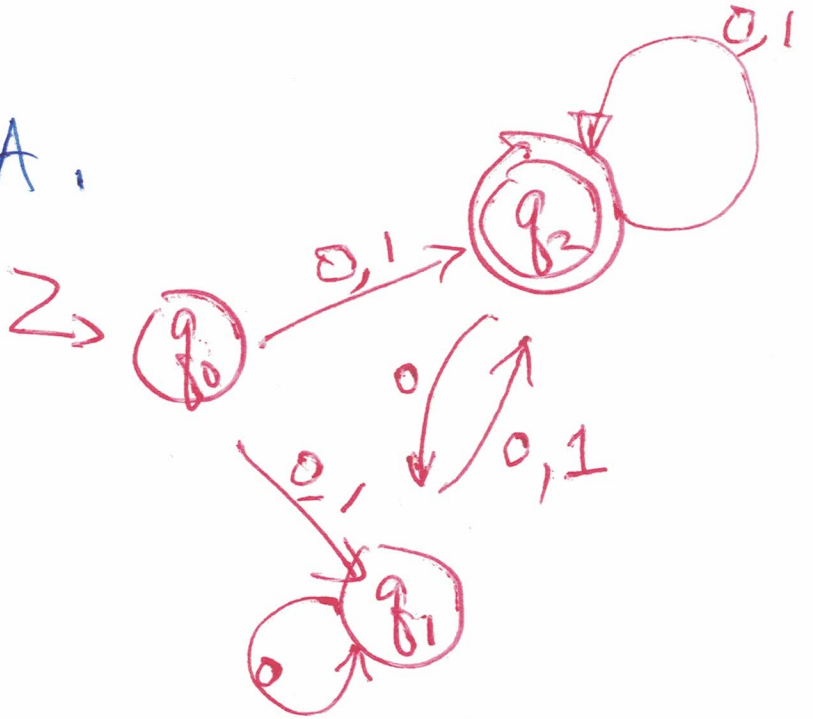
   (NFA $\rightarrow$ NFA).

Exaple of Λ- Elemination (Λ-NFA $\Rightarrow$ NFA).

Given: the following Λ-NFA



Want: Convert it to NFA.



Ask yourself: from q_0, we read 0, where can we go?

$\Lambda \ldots \Lambda 0 \Lambda \ldots \Lambda$

I know:

$$\Lambda\text{-NFA} \xrightarrow[\text{Λ-Elimination.}]{} \text{NFA} \xrightarrow[\text{subset const.}]{} \text{DFA}.$$

Kleene theorem. For each language L,

L is reg. iff L is accepted by a FA.

// FA $\equiv$ DFA, NFA, Λ-NFA.

//

//

//

Proof has two parts:

Part 1: Each reg. language is accepted by a FA. (language-to-automaton construction)

Part 2: Each FA accepts a reg language (automaton-to-language construction)

Practical Implications of Kleene Theorem.

Finite Automata
(programs that use
fixed amount of memory)

||

Code

Regular languages.
(a simplest class of languages)

floor1 : (floor2 + floor3)*

||

Requirements

automatic

Caution:

NFA $\longrightarrow$ DFA

n states $\qquad$ 2^n states.